Three compatible decoder reductions

A 209-state candidate for Con(PA)

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Abstract

The three supplied 211-state proposals delete different native states from a common 212-state ancestor. Their final tables cannot be combined by simply appending the three patches: their decoder assignments conflict. We construct a new source association that makes all three reductions simultaneously valid. The resulting binary, one-tape machine has exactly **209 working states**, including blank initialization, and 418 transition rows. The new local reduction has a complete ten-case proof over arbitrary exterior tapes. Independent source and startup audits pass, and all 5,084 primitive contracts are rederived over arbitrary natural register values and finite spare lists. The consistency interpretation composes these results with the supplied numerical-calculus and Theory U calibration. This is a mathematical construction with executable inductive evidence; it is not a new Lean, Rocq, MM0, or internal-PA proof object.

1 Result and exact machine conventions

The primary artifact is `machine/u209.tm`. It has working states $0, \ldots, 208$, initial state 0, alphabet $\{0, 1\}$, a single tape indexed by $\mathbb{Z}$, initial head position 0, and an initially all-zero tape. Every working-state/symbol pair has exactly one transition. All moves are left or right. Label 209 is the distinct halt label, not another working state. Initialization is included in the 209-state count.

A row `q b w D r` means: in state q reading b, write w, move one cell in direction D, and enter r. A transition into the halt label performs its write and move before halting. The header `#! start 0` is not a transition. The complete table appears in Appendix A. The native table retains the inherited sparse labels and halt 254; these labels are not state counts. `machine/state_map.json` supplies the exact bijective renaming.

All 209 working states are reachable in the directed transition graph. This does not assert that all occur in the particular blank-tape execution. The construction deletes native states **180, 181, and 218** from the common 212-state parent. Thus it improves every supplied 211-state candidate by two working states, while leaving the ordered source program and its arithmetic proof search unchanged.

Property	Delivered result
Working states / literal binary rows	209 / 418
Ordered primitive occurrences / distinct source subtrees	3,909 / 197
Guarded primitive contracts	5,084
Preserved nonhalting source control-flow edges	5,083
Changed source addresses, relative to U212	3,096
Maximum source depth / consecutive-zero run	42 / 6
Exact blank startup	48,140 transitions
Direct local simulation cases / rejected corruptions	10 / 12
Symbolic inference applications	1,250,049
Literal transition rows covered	418 / 418

The intended endpoint, with the inherited logical calibration made explicit, is

$$\mathsf{PA} \vdash \neg\,\mathrm{Halt}(M_{209}, \mathrm{blank}) \iff \mathrm{Con}(\mathsf{U}) \iff \mathrm{Con}(\mathsf{PA}). \tag{1}$$

The newly checked machine obligations and the inherited proof-search/theory obligations are separated in Sections 8–10. Equation (1) is the conclusion of their mathematical composition, not the output of a proof-assistant kernel run in this investigation. Actual nonhalting follows under the assumption that PA is consistent.

2 Pinned inputs and comparison of the three proposals

The review is pinned to ProveIt commit

`da4ab9dd68ccfc47ac852118601eaf1fdae53795`.

At that snapshot, `docs/incoming` contains the three named U211 archives and `pa_unary_power_cover.zip`. Each download was checked against its repository byte length and Git blob SHA-1. Their SHA-256 values, exact URLs, and the machine/source comparisons are retained in `comparison/`. The original four archives are included intact under `reference/`.

The three 211-state tables are genuinely different. Each derives from the same native U212 table, whose SHA-256 is

`386ce16259d4c97e4af31b3ba494feaff11a0a1a066e0235dddc154b4cea38c4`.

The source comparison independently confirms that all three have the same ordered primitive effects and the same complete ordinal control-flow graph as this parent. Their common 3,909-leaf program has 197 distinct subtrees.

Contextual changes 16 retained rows, Research 19, and Rotated 19, each deleting two additional rows. Contextual and Research have maximum source depth 42; Rotated has depth 41. Every proposal retains maximum zero run six, which is compatible with the inherited seven-zero origin detector. These are engineering differences, not differences in the strength of their logical endpoint.

For a single reduction to formalize first, Rotated's two-cell deletion is the cleanest local lemma. Research supplies a useful independent selector whose helpers can coexist with that deletion. Contextual supplies the third saving, but its layout competes with Rotated for native 163 and its non-decoder entry requires attention. The new result resolves those physical allocation conflicts.

Archive	Native deletion	Addresses changed	Construction and useful distinction
Contextual	$180 \mapsto 163$	549	Six source rotations, a repair of the non-decoder entry 171/0, and a preceding-bit selector through 211. Smallest changed source region; the entry repair must be retained.
Research	$181 \mapsto 220$	936	Nine rotations expose two selectors with opposite preceding bits and a common read-1 branch. Existing helper 129 distinguishes their read-0 branches. The incoming-entry guards are explicit.
Rotated	$218 \mapsto 157$	3,096	Eleven rotations arrange that every entry to 218 writes zero immediately left of the head. Existing 157 implements either 218 transition in three steps. This has the simplest isolated deletion proof.

Table 1: The supplied 211-state proposals, compared against their common U212. The native table hashes are given in the comparison receipt.

All three packages describe ordinary mathematical reductions with executable symbolic induction; none supplies a newly checked end-to-end kernel theorem for its U211 table. The K215/Con(ZFC) results cited by the user concern another literal machine and another consistency predicate [6, 7]. The verified m254 development likewise provides a relevant layered verification model, not a certificate for this changed table [9].

3 Theory U and the calculus actually executed

The supplied theory report uses classical equality-free first-order logic over a nonempty domain, with membership as its only nonlogical symbol. Its schema is

$$\forall a\, \exists b\, [\forall x\, (x \in b \to \varphi(x)) \ \wedge \ \forall x\, (x \subseteq a \to (\varphi(x) \to x \in b))]\,, \tag{2}$$

where $x \subseteq a$ abbreviates $\forall z\, (z \in x \to z \in a)$ and φ has no free variable other than x. Bound names are freshened. In particular, a is not an extra free matrix parameter. This is weak covering, not ambient extensionality and not an asserted exact power-set operation.

For the upper interpretation, PA interprets $x \in a$ as the corresponding binary-bit relation. Bit inclusion implies a numerical bound $x \leq a$. Formula-specific finite-comprehension induction then constructs a cover satisfying (2). This uses a fixed formula at a time, not a uniform truth predicate. The lower interpretation in the supplied report develops numerical representatives modulo coextensionality, local predecessor induction, least-failure comprehension, numerical successors and powers, and graph/history codes for arithmetic. Saturation is used where representative equality must respect membership. The parameter restrictions are handled inside legal unary matrices. The intended result is uniform finite-refutation translation provable in PA, which is stronger information than a bare external assertion that the two theories happen to be equiconsistent [5].

The machine does *not* newly compile the positive-reference stream described elsewhere in that theory packet. It executes the inherited numerical calculus $\mathcal{D}_U$: formula truth and falsity codes 0 and 1, logical rows, a dominated propositional row, modus ponens, generalization, and a gated covering bundle. Its machine-facing set theory allows matrix parameters and uses global irreflexivity. The inherited calibration relates that theory, the numerical calculus, unary U, and PA by finite-refutation translations. The literal axiom sets of those presentations are not being identified.

The practical consequence is that a smaller decoder can preserve the numerical proof search without re-proving a different set theory. Conversely, verifying this decoder cannot replace the logical calibration. The original packages and their retained ancestry are included so those separate arguments remain available for review.

4 Why the three deletions do not simply stack

In Research's useful allocation, helper 129 must distinguish two contexts:

$$129 = (158, 204), \qquad 158/0 \to 218, \qquad 204/0 \to 147.$$

Here (a, b) denotes a pure decoder with read-0 child a, read-1 child b, unchanged symbols, and right moves. The states 220 and 181 must receive preceding bits 0 and 1, respectively, and share read-1 child 176.

Rotated's deletion needs *every* incoming transition to 218 to move right after writing zero. It achieves this partly by changing the source occurrence in which 218 formerly occupied a read-1 child. In its allocation, native 163 serves a tail of the form $(218, 116)$.

Contextual instead needs native $163 = (90, 211)$, entered after a written zero. Native $180 = (152, 211)$ must be entered after a written one. It can then share 163 through helper 211. Thus copying the final Contextual rows over Rotated's final table would destroy the tail through 116; preserving the wrong association would also reintroduce an entry to 218 with the wrong preceding bit.

The new step is to keep Research's selector, move the continuation through the closed source sequence, and use native **215** for the tail $(218, 116)$. This frees native 163 for $(90, 211)$ while keeping all entrances to 218 tagged zero. The local savings are inherited ideas; their simultaneous realization in one source and one literal table is the new construction.

5 The compatible source layout

The deterministic builder starts with the frozen U212 table and source. It first applies the nine Research rotations, then eight rotations transporting the closed continuation at 116. These yield the independently checked 210-state intermediate if only the first two folds are applied. For U209, a further closed-factor reassociation within native 174 and the three external Contextual rotations establish all three entry invariants.

For a pure decoder p with inner child q, side 0 denotes $((A, B), C) \mapsto (A, (B, C))$, and side 1 the reverse. Each ordinary rotation checks the literal rows, unique incoming entry to the inner decoder, matching source occurrences, and closure of the three factors. Closure means no halt and no branch escaping the factor except normal sequential continuation. The builder also compares every primitive and every resolved successor after each stage. The complete recipe is recorded in the construction receipt.

Step	Outer	Inner	Side	Step	Outer	Inner	Side
1	215	204	1	11	203	223	0
2	217	181	1	12	223	188	0
3	185	217	0	13	188	177	0
4	174	185	1	14	177	89	0
5	129	158	0	15	89	174	0
6	217	215	1	16	174	217	0
7	204	214	0	17	217	163	0
8	215	129	0	18	206	199	1
9	216	220	1	19	205	206	1
10	162	203	0	20	7	205	1

Between steps 17 and 18, the 720-primitive region rooted at native 174 is reassociated as follows. Its unchanged closed factors, in their exact order, are

$$98,\ 218,\ 178,\ 147,\ 146,\ 218,\ 176,\ 90,\ 211,\ 218,\ 116. \tag{3}$$

Their lengths are $26, 56, 78, 52, 108, 56, 80, 27, 103, 56, 78$, summing to 720. Repeated labels denote identical factors. None of the internal states except the region root has an external incoming edge. The final internal assignment is

State	Read-0 child	Read-1 child	State	Read-0 child	Read-1 child
174	98	129	216	220	217
129	158	204	220	218	176
158	218	178	217	163	215
204	147	214	163	90	211
214	146	216	215	218	116

All rows in this display preserve the read symbol and move right. This table, rather than a heuristic search, determines the new association exactly.

The external rotations re-associate

$$(86, (164, (186, (180, 166)))) \quad \longmapsto \quad ((86, (164, (186, 180))), 166).$$

They arrange the two incoming transitions to 180 to write one without creating a seven-zero source address. Because native 7 also has a non-decoder use, the builder then applies the required repair

$$171/0 : 0\,\mathrm{L}\,7 \quad \longmapsto \quad 0\,\mathrm{L}\,205. \tag{4}$$

For completeness, this repair has a two-cell proof. If the current bit is zero and the bit immediately left is t, the old and new paths are

t	old	new	common endpoint
0	$171 \to 7 \to 86$	$171 \to 205 \to 86$	$(86, h)$
1	$171 \to 7 \to 205 \to 164$	$171 \to 205 \to 206 \to 164$	$(164, h+1)$.

The last transition in the second case reads the zero just written by 171. Both paths preserve the whole tape, take two or three steps as displayed, and read no exterior cell. The independent source/startup checker replays them against the actual parent and final tables.

Call the resulting 212-state table R_{212}. It is included as `machine/prefold212.native.tm`. Source reassociation changes instruction addresses. Therefore the connection between U212 and this representation uses source semantics and canonical frames; an arbitrary-common-tape equivalence is not asserted for that part of the construction.

6 Three compatible folds and their entry invariants

From R_{212}, delete 181 into 220, delete 218 into 157, and delete 180 into 163. Redirect every incoming target accordingly. Set the surviving rows

$$220/0 = 0\,\mathrm{L}\,129, \qquad 163/0 = 0\,\mathrm{L}\,211.$$

All other surviving rows are the rows of R_{212} with their targets renamed by

$$\rho(181) = 220, \qquad \rho(218) = 157, \qquad \rho(180) = 163, \qquad \rho(q) = q \quad \text{otherwise.} \tag{5}$$

This produces 209 working states and 418 rows. The native final SHA-256 is

`d0a155d3c683789f3bf5327c0cfb2a0cfd78076af8f1654d4ca426ccbb604d16`.

The complete incoming inventories in R_{212} establish the necessary tags:

Old state	Required bit at $h-1$	All incoming rows in R_{212}
220	0	$216/0 : 0\,\mathrm{R}\,220$
181	1	$185/1 : 1\,\mathrm{R}\,181$
218	0	$158/0, 211/0, 215/0, 220/0$, all $0\,\mathrm{R}\,218$
163	0	$217/0 : 0\,\mathrm{R}\,163$
180	1	$132/1, 199/1$, both $1\,\mathrm{R}\,180$

None is the initial state. The previous transition writes the tag, so these invariants hold on every initial binary tape. They require no hypothesis about the numerical proof search, and no guessed unreachable branch.

Lemma 6.1 (Ten exact local identities)**.** *For each old state in* $\{163, 180, 181, 218, 220\}$ *and each current bit, assume only the indicated bit at* $h-1$*. One transition of* R_{212} *is simulated by the following path of* M_{209}*, with control renamed by* ρ*, identical final head position, and identical whole tape.*

Old state	Tag	Read	New control path, including endpoint	Steps
163	0	0	$163 \to 211 \to 157 \to 67 \to 151 \to 90$	5
163	0	1	$163 \to 211$	1
180	1	0	$163 \to 211 \to 184 \to 152$	3
180	1	1	$163 \to 211$	1
181	1	0	$220 \to 129 \to 204 \to 147$	3
181	1	1	$220 \to 176$	1
218	0	0	$157 \to 67 \to 151 \to 90$	3
218	0	1	$157 \to 148 \to 186 \to 86$	3
220	0	0	$220 \to 129 \to 158 \to 157$	3
220	0	1	$220 \to 176$	1

Proof. Place the old head at 0 and specify the two cells -1 and 0. Follow the literal rows in the final table. Every path rewrites the symbol already present and finishes with head 1; no path reads any other cell or passes through halt. For example, the five-step row reads, in order, cells $0, -1, 0, -1, 0$, all zero, and finishes at state 90. The state-220/read-0 path ends at 157, which is $\rho(218)$, not at 90; the old transition only enters 218. This distinction is essential when composing the folds. The other eight cases follow the displayed rows in the same way. The executable checker represents the exterior as unspecified and rejects any read outside the two known cells, so the checks are finite proofs of identities valid for every exterior tape. □

Theorem 6.2 (Direct whole-tape simulation)**.** *From state 0 on every common initial tape, R_{212} and M_{209} halt equivalently. At synchronized configurations their tapes and head positions agree and their control states are related by ρ. One old transition corresponds to one, three, or five new transitions.*

Proof. Relate (q, h, T) to $(\rho(q), h, T)$, imposing the tag shown above only for the five special old states. The relation holds initially because $q = 0$. For any of the other 207 old states, both rows have identical writes and moves, and their targets commute with ρ. These are 414 ordinary-row checks. If such a row enters a special state, its complete incoming inventory supplies the required new tag. For a special state, Lemma 6.1 supplies the corresponding positive finite block. The only special endpoint among those paths is old 218 after old 220/0; the path leaves its predecessor cell zero, as required.

Induction on the old step count now constructs the synchronized blocks. Both tables are deterministic and have no undefined working transition. An old halting computation therefore yields a new halting computation. For the converse, suppose the new computation halts after t steps and the old one does not halt during its first t transitions. The block construction then yields at least t new transitions without passing through halt, except possibly at a synchronized endpoint. Such an endpoint is halt exactly when the corresponding old endpoint is halt, since ρ preserves the halt label and does not send a working state to halt. This contradicts the proposed new halting time. The argument uses only finite induction bounded by t.

Original visits to native 157 and 211 are ordinary states in this relation. They do not inherit the predecessor restrictions of the deleted states whose behavior they additionally implement. Thus sharing the helpers introduces no unstated global tape invariant. □

For a halting old run, let N_3 count its transitions from 218 with either read symbol, or from 180, 181, or 220 reading zero. Let N_5 count its transitions from 163 reading zero. The sets are disjoint. The exact clock law is

$$T_{209} = T_{R_{212}} + 2N_3 + 4N_5, \qquad T_{R_{212}} \le T_{209} \le 5T_{R_{212}}. \tag{6}$$

This concerns the final reassociated 212-state table R_{212}. It is not a runtime comparison with the original U212 or any supplied U211 representation.

7 Source preservation and the independent prefix audit

The source is a full binary tree with leaves CI, CD, CB, CC, CH: increment, guarded decrement, forward branch, backward branch, and halt. Their occurrence counts are respectively 1,558, 1,175, 80, 1,095, and 1. A source state is a leaf ordinal together with nine natural register values. Increment and positive decrement continue to the next leaf; a zero decrement or a branch uses the appropriate ancestor cut. The resolved ordinal successor is the semantic object that must be preserved.

For a forward cut at the kth enclosing left ancestor, the successor is the first leaf after that ancestor's interval. For a backward cut at the kth enclosing right ancestor, it is the first leaf of that ancestor's interval. These definitions yield 5,083 nonhalting successor arrows; adding the halt case gives 5,084 guarded primitive cases. Checking only the leaf order would not establish this control-flow property.

Proposition 7.1 (Identity on source executions)**.** *The final source and the common U212 source have identical opcodes, operands, and ordered successor ordinals at every leaf. From a common ordinal and store, their abstract executions have the same ordinal and store at every source step, including the same halting step if one exists.*

Proof. The builder checks the control graph after each rotation and after the closed 720-primitive reassociation. It uses both interval cuts and independently computed address arithmetic. A separate checker reads only the two delivered source lists and literal tables, reconstructs their prefix trees and intervals, and compares every edge. The finite equality checked is precisely the premise needed for induction on source steps. Split a decrement step according to whether the tested register is zero; every other instruction has its prescribed unique effect or halts. The induction gives the conclusion. □

The independent checker also follows every one of the 3,909 primitive entry prefixes in the original, final pre-fold, and final 209-state tables. It knows only the proper prefix and its left sentinel; the exterior is opaque. Every proper prefix preserves the tape and reaches an identical terminal primitive action in the three tables. This checks literal decoder binding in addition to the abstract source graph. It does not replace the complete primitive runtime contracts below, because the interpreter can re-enter decoder states during arithmetic and address updates.

8 Universal primitive contracts

For $v \in \mathbb{N}$, put $B(v) = 0\,1^{v+1}$ and $C(v) = 1\,0\,1^{v}$. For a finite list A, the words $B(A)$ and $C(A)$ are concatenations of the corresponding blocks; the empty list gives the empty word. At a primitive boundary the control is native 84, with the head at the first bit of the source address p. The whole tape is represented by

$$\cdots 0\,1\;p\;0^{56+g-|p|}\,1\;B(v_0)\cdots B(v_8)\,B(a_0)\,B(A)\;0\cdots . \tag{7}$$

Here g, a_0 are arbitrary naturals, A is an arbitrary finite list of naturals including the empty list, and the nine logical register values are arbitrary naturals. The physical permutation is

$$\pi = (6, 2, 0, 4, 3, 7, 8, 5, 1), \qquad v_{\pi(i)} = x_i.$$

Thus the full spare list is nonempty, and both infinite zero exteriors are part of the frame. The maximum address length 42 leaves at least 14 padding zeros at minimum header distance 56.

The retained symbolic engine works with thirteen affine coordinates: a constant, the extra gap, nine primary register values, the first spare value, and the actual total encoded width of the remaining spare list. It splits every decrement into the exhaustive cases $x = 0$ and $x = u + 1$. This is not a finite sampling of register valuations.

Its proof rules have elementary mathematical justifications:

1. **Literal step.** Apply a row of the final table. A non-loop literal read requires a run length with constant term at least one and nonnegative coefficients. Thus the cell exists for every represented valuation. Unknown or opaque cells cannot be read by this rule.
2. **Unary-run induction.** A self-loop moving in the current traversal direction can consume an entire homogeneous run. Induction on its arbitrary natural length proves the repeated transition and its exact affine time and displacement, including length zero.
3. **Finite-list induction.** Four one-block laws are derived anew from the final table, with an arbitrary natural block value and opaque surrounding contexts. They traverse a block in exactly $v + 2$ steps, returning to the traversal state. The empty-list case and concatenation prove the corresponding laws for every finite spare tail.
4. **Word identities.** Adjacent runs are joined, zero-length runs removed, and a zero exterior absorbs adjacent finite zeros. The additional identity $C(A)1 = 1B(A)$, and its reversal, follow by expanding one block and induction on the list. These identities preserve the whole tape word.

For each of the 5,083 nonhalting contracts, the engine checks the exact resulting control, both tape sides, head displacement, source-address update, register changes, and required spare-block changes. The remaining CH contract proves that the machine reaches halt in finite positive time. All 5,084 durations are affine with nonnegative coefficients and a strictly positive constant. The final run uses **1,250,049 symbolic inference applications**, covers all **418** literal rows, and has minimum duration constant **5**. The one-block lemmas are derived from the same final table. Coverage is a useful diagnostic; the quantified contracts and the induction rules are the substantive evidence.

The symbolic engine's inference-count cutoff limits certificate construction, not the register values certified by a successful derivation. The engine uses Python assertions and the full replay deliberately rejects optimized Python. Its code is retained with provenance, and its proof rules are ordinary mathematical obligations. It is not a proof-assistant kernel.

The finite source and local-fold checkers provide independent cross-checks: they import neither the builder nor the symbolic engine, and their explicit checks remain active under `python3 -O`. In particular, the new five-step path through helper 157 is checked directly rather than inferred from a promise that independently discovered patches should compose.

9 Exact initialization and the inherited protected suffix

The final table's first 48,140 blank-tape transitions agree exactly with the common U212 trace. No edited row is visited. The endpoint is native 84 with head -2, header separator at 54, and source ordinal 136, whose address is `001100110101010`. The logical registers are

$$(1, 0, 13, 0, 0, 1, 0, 0, 1), \tag{8}$$

and there are 43 spare registers, each equal to 1. The independently reconstructed complete tape has 120 one-cells and is zero elsewhere. Thus initialization is an exact finite computation establishing frame (7), not an assumed input encoding.

The independent source audit additionally executes 141 source steps from ordinal 106 with store $(1, 0, 4, 0, 0, 1, 0, 0, 1)$, reaching precisely ordinal 136 and store (8). This confirms the interface to the inherited suffix argument; it does not by itself prove that argument.

The retained calibration is delicate because the literal startup enters a suffix of the axiom builder rather than a clean proof-enumerator root. That suffix builds nine valid theorem heads above an opaque numerical tail 4. The initial cache 1 is an arithmetic selector parameter, not an assumed theorem. The tail must remain protected: in the inherited pairing convention it contains codes that would be unsafe if exposed. The supplied scheduler proof gives a maximum prefix debt of two, leaving margin seven beneath the nine heads; an explicit ghost length handles the zero-pair alias. Soundness therefore requires that protection, and completeness uses the strong compiler above arbitrary residual stacks together with cofinal successful candidate indices.

The present source transformations preserve every source ordinal and effect, and their literal startup reaches the same suffix state. They therefore preserve these calibration obligations; they neither remove the tail nor replace the argument with a claim that a long finite run is evidence of consistency.

10 The precise consistency conclusion

There are three levels of result.

First, Theorem 6.2 is a direct local theorem about two fully explicit finite tables. It holds on arbitrary common initial tapes from state 0. For encoded finite-support tapes, its finite-trace proof is formalizable in PA.

Second, the universal primitive contracts and exact initialization connect the final literal machine to the fixed abstract source execution starting at ordinal 136 and store (8). Every nonhalting source step returns after finite positive time to the canonical frame, including a nonempty spare list; the CH instruction reaches literal halt. Induction therefore transports each finite source execution to a literal execution. Conversely, the same positive block decomposition precludes a literal halt except at the corresponding source halt. This is the machine-to-source bridge for the final table. Source reassociation is justified at this level, with its changed address encoding.

Third, the supplied source-program, numerical-calculus, and theory arguments identify that source's halting with a finite $\mathcal{D}_U$ refutation and identify $\mathcal{D}_U$ refutation with PA refutation. Explicitly, the inherited obligations are

$$\mathsf{PA} \vdash \mathrm{Halt}(\text{specified source suffix}) \leftrightarrow \mathrm{Ref}(\mathcal{D}_U), \tag{9}$$

$$\mathsf{PA} \vdash \mathrm{Ref}(\mathcal{D}_U) \leftrightarrow \mathrm{Ref}(\mathsf{U}) \leftrightarrow \mathrm{Ref}(\mathsf{PA}). \tag{10}$$

They include the protected-tail and cofinality results of the preceding section and uniform finite-proof translations, not an appeal to semantic truth of PA. Combining the new machine bridge with (9)–(10), then negating the refutation predicates, gives (1).

Accordingly, the delivered result is a **209-state Con(PA) candidate with a complete new machine reduction and explicitly inherited calibration**. The research checked the theory argument for the relevant parameter, saturation, and coding issues and found no concrete contradiction with this use. It did not generate a new proof-assistant object for the inherited logical theorems. Readers who wish to review the machine compression independently can stop at the first two levels; readers assessing the Con(PA) claim must also review the retained calibration. The separate K215 theorem is not substituted for either obligation.

11 Negative controls and limits of the search

The local checker rejects twelve explicit corruptions: wrong predecessor tags for the five special states, helper writes that alter the required tag or current bit, a wrong target for the 218 quotient, wrong directions at the two merged selectors, and an incorrect 181 redirection. Some controls change a helper in both old and new tables, so ordinary-row correspondence alone would still pass; the two-cell proof must detect the error. The independent audit passes with assertions disabled, while the full symbolic replay correctly refuses that mode.

A bounded exploratory screen found no additional one-row contextual merge on the examined U211 and U210 tables within its finite-window and path-length limits. This is search information only. No lower bound on the number of states follows, and no global minimality or priority claim is made. The useful discovery was to solve the shared layout constraints before attempting the three eliminations.

Two tempting theory changes are also excluded for the unchanged covering emitter. Its freshness gate requires the quantified label c to differ from the output label a, also used as the matrix code, and the source label b. Removing only $c \neq a$ permits the truth-matrix instance $(a, b, c) = (0, 1, 0)$; under irreflexivity the covering condition demands self-membership. Removing only $c \neq b$ permits $(0, 1, 1)$ and demands a universal object, again contradicting irreflexivity. These are exact small-code obstructions to those gate deletions, not an argument against a redesigned calculus with a different soundness proof.

The local clock law does not establish that U209 is faster than the supplied machines. The source has the same 3,909 operations and 197 distinct subtrees; the gain is physical control-state reuse. The interpreter's changed addresses and extra local steps can alter execution cost. State count is the optimized quantity here.

12 Further research questions

1. **Can the exact 209-state endpoint be checked in a small kernel?** Formalize the ten-case quotient lemma and its incoming-tag criterion first, then import or formalize the symbolic run/list rules, startup frame, source compiler, numerical calculus, and U/PA proof translations. The precise table hash should occur in the final endpoint statement or its verified data closure.
2. **Can source layout be synthesized subject to several helper contracts?** The successful construction preserves three local interfaces at once. A search over associations and decoder assignments could retain those interfaces as hard constraints, along with exact source control flow, external entry behavior, address bounds, and a final universal-runtime gate. This is more targeted than independently optimizing one deletion and trying to append others afterward.
3. **Can longer predecessor contexts support another deletion?** Two known neighboring bits, or a locally reconstructible control history, may allow a further state to share an existing helper. A candidate must come with an inductive entry invariant and an exact finite path that preserves the whole tape. A successful sample trace is not that invariant.
4. **Can source-address coding reduce the origin detector safely?** The current source has legitimate runs of six zeros. A shorter detector needs new source layouts and control over temporary address-update words, not just a maximum-run count on final instruction addresses. Startup and every primitive must be rechecked for any such change.
5. **Would compiling the unary theory's positive stream pay for its different enumeration machinery?** The current result preserves $\mathcal{D}_U$. A new compiler for the separate dyadic stream is a distinct research route requiring its own finite-refutation proof and measured literal state count. One schema alone does not determine that cost.

13 Reproduction and archive contents

Python 3.10 or later and its standard library suffice for the machine evidence. From the extracted archive directory, run

```
python3 reproduce.py
```

This verifies the frozen manifest and inputs, reconstructs the machine and source into `build/`, compares them with the delivered files, replays the independent local and source/startup audits, and reruns every universal primitive contract on the final table. No network, discovery search, external solver, or proof-assistant installation is used. The replay compares the newly derived case data with the retained data and checks row coverage and positive duration.

```
python3 reproduce.py --quick
```

The quick mode explicitly skips the universal runtime replay. It is a useful integrity/construction/local check, not a replacement for the full command. The symbolic engine requires assertions; do not use `-O` or `PYTHONOPTIMIZE` for the full replay. Regenerated receipts are written to `build/evidence/`; the released evidence is preserved.

The supplied PDF can be rebuilt with a normal TeX installation:

```
pdflatex -interaction=nonstopmode -halt-on-error report.tex
pdflatex -interaction=nonstopmode -halt-on-error report.tex
```

Rebuilding the PDF can change its metadata and hash. Check the supplied manifest before rebuilding it, or compile a separate copy. The mathematical tables and source bytes are fixed outputs independent of TeX metadata.

Path	Contents
`report.tex`, `report.pdf`	Report, complete proofs of the new local reduction, calibration boundary, and dense transition appendix.
`machine/`	Final native/dense U209, exact state renaming, source sites, pre-fold intermediate tables, and construction receipt.
`code/`	Deterministic builder, independent local and source/startup auditors, universal-runtime wrapper, and retained symbolic engine.
`inputs/`	Frozen common U212 table/source and construction provenance.
`evidence/`	Source/startup, ten-case local proof, rejected corruptions, universal case data, and release replay receipts.
`comparison/`	Pinned GitHub input identities and independently computed three-proposal comparison.
`reference/`	The three original U211 ZIPs and the unary-theory ZIP, unchanged, preserving their reports and inherited proof dependencies.
`NOTICE.txt`, `licenses/`	Attribution and inherited licenses.
`MANIFEST.json`	Byte length and SHA-256 of every frozen payload file.

A Complete dense transition table

Working labels are 0–208 and halt is 209. Each cell contains *write, direction, next state*. Read columns 0 and 1 give both rows for each state; therefore the appendix specifies all 418 transitions. The separate `machine/u209.tm` is the primary machine-readable form.

State	Read 0	Read 1	State	Read 0	Read 1
0	1 R 92	1 R 0	35	1 R 35	1 L 1
1	0 L 1	1 L 2	36	0 R 50	1 R 36
2	0 L 3	1 L 2	37	0 R 17	1 R 111
3	0 L 5	1 L 2	38	0 R 13	1 R 52
4	1 R 6	0 L 4	39	0 L 11	0 R 36
5	0 L 9	1 L 2	40	0 L 4	1 L 53
6	0 R 6	0 R 10	41	0 R 24	0 R 25
7	0 R 190	1 R 159	42	0 R 13	0 L 16
8	0 R 14	1 R 13	43	0 R 55	1 R 56
9	0 L 15	1 L 2	44	0 R 57	1 R 58
10	0 L 16	0 R 10	45	0 R 24	0 R 54
11	0 L 26	1 R 128	46	0 R 60	1 R 61
12	0 R 19	1 R 20	47	0 R 62	0 L 16
13	0 L 89	1 L 31	48	0 R 25	1 R 48
14	0 L 11	0 R 22	49	0 L 35	0 R 63
15	0 L 23	1 L 2	50	0 R 16	1 R 50

Continued on the next page

State	Read 0	Read 1	State	Read 0	Read 1
16	1 L 1	0 L 1	51	0 R 147	1 R 143
17	0 R 24	0 R 32	52	0 R 67	1 R 68
18	0 R 98	1 R 113	53	0 L 40	1 L 53
19	0 R 28	1 R 29	54	0 R 22	1 R 54
20	0 R 30	1 R 31	55	0 L 11	0 R 50
21	0 R 24	0 R 59	56	0 R 60	0 L 16
22	0 R 33	1 R 22	57	0 R 69	1 R 70
23	0 L 34	1 L 2	58	0 R 61	1 R 71
24	0 L 35	0 R 0	59	0 R 54	1 R 59
25	0 R 36	1 R 25	60	0 R 49	0 R 25
26	0 R 37	1 R 38	61	0 L 28	1 L 72
27	0 R 39	0 L 40	62	0 L 117	1 L 30
28	0 R 41	1 R 42	63	0 R 73	1 R 63
29	0 R 43	1 R 44	64	0 R 97	1 R 150
30	0 R 45	1 R 46	65	1 R 185	1 L 53
31	0 R 21	1 R 47	66	0 R 144	1 R 145
32	0 R 48	1 R 32	67	0 R 76	1 R 209
33	0 R 32	1 R 33	68	0 L 197	1 R 31
34	0 L 74	1 L 2	69	0 R 41	1 R 62
70	0 L 89	1 R 78	105	0 R 21	0 L 10
71	0 R 55	1 L 79	106	0 R 61	1 R 42
72	0 R 80	1 R 81	107	0 R 121	1 R 8
73	1 R 0	1 R 63	108	0 R 101	1 R 122
74	0 R 74	1 R 82	109	0 L 208	0 L 95
75	0 L 145	1 L 141	110	0 R 124	1 R 104
76	0 R 85	1 R 86	111	0 R 115	1 L 72
77	0 R 20	1 R 38	112	0 R 165	1 R 181
78	0 R 89	1 L 132	113	0 R 160	1 R 184
79	0 L 108	1 R 56	114	0 R 148	1 R 146
80	0 R 49	0 R 33	115	0 R 11	1 R 129
81	0 R 91	0 L 42	116	0 L 131	0 L 1
82	0 R 178	1 R 77	117	0 R 39	1 R 72
83	0 R 96	1 R 12	118	0 R 61	1 R 132
84	0 R 88	1 R 97	119	0 R 61	1 R 133
85	0 R 207	1 R 21	120	0 R 70	0 L 16
86	0 R 116	1 R 100	121	0 R 14	1 R 132
87	0 R 174	1 R 167	122	0 R 21	1 R 134
88	0 R 102	1 R 103	123	0 R 151	1 R 189
89	0 R 80	1 R 79	124	0 R 137	1 L 20
90	0 R 101	1 R 196	125	0 R 130	1 R 180
91	0 R 105	1 R 106	126	0 R 204	1 R 156
92	0 R 65	0 R 0	127	0 R 75	1 R 176
93	0 R 27	1 R 107	128	0 R 75	1 R 206
94	0 R 183	0 L 208	129	0 R 203	1 R 157
95	1 L 109	1 L 95	130	0 R 12	1 R 141
96	0 R 19	1 R 110	131	0 R 30	1 R 69
97	0 R 60	1 R 60	132	0 R 39	1 L 124
98	0 R 68	1 R 135	133	0 R 153	0 L 16
99	0 R 93	1 R 108	134	0 R 154	0 L 16
100	0 R 105	0 L 208	135	0 R 88	1 R 136
101	0 R 117	1 R 118	136	0 R 197	1 R 140
102	0 R 71	1 R 58	137	0 R 24	0 R 48
103	0 R 57	1 R 119	138	0 R 99	1 R 37

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State	Read 0	Read 1	State	Read 0	Read 1
104	0 R 21	1 R 120	139	0 R 172	1 R 148
140	0 R 96	1 R 159	175	0 R 140	1 R 84
141	0 R 175	1 R 166	176	0 R 175	1 R 75
142	0 R 12	1 R 159	177	0 R 126	1 R 169
143	0 R 88	1 L 166	178	0 R 138	1 R 116
144	0 R 88	1 R 158	179	0 R 97	1 R 51
145	0 L 152	1 L 163	180	0 R 168	1 R 51
146	0 R 84	1 R 88	181	0 R 17	1 L 131
147	0 R 159	1 R 12	182	0 R 143	1 R 149
148	0 R 83	1 R 143	183	0 R 76	1 R 116
149	0 R 144	1 R 155	184	0 R 106	1 R 124
150	0 L 66	1 L 141	185	1 L 40	1 L 109
151	0 R 150	1 R 170	186	0 R 175	1 R 156
152	0 R 159	1 R 114	187	0 R 182	1 R 183
153	0 R 97	1 R 97	188	0 R 66	1 R 205
154	0 R 49	0 R 32	189	0 R 140	1 R 198
155	0 R 192	1 R 188	190	0 R 84	1 R 191
156	0 L 195	1 R 195	191	0 R 157	1 R 186
157	0 L 187	1 R 96	192	0 R 64	1 R 7
158	0 R 168	1 R 179	193	0 R 150	1 R 127
159	0 R 19	1 R 171	194	0 R 18	1 R 94
160	0 R 164	1 R 187	195	0 R 150	1 R 173
161	0 R 159	1 R 96	196	0 R 104	1 R 96
162	0 R 125	1 R 164	197	0 R 90	1 R 88
163	0 R 97	1 R 162	198	0 R 139	1 R 200
164	0 L 190	1 R 142	199	0 R 150	1 R 112
165	0 L 66	1 R 173	200	0 R 202	1 R 201
166	0 R 146	1 R 172	201	0 R 156	1 R 199
167	0 R 96	1 R 123	202	0 L 123	1 R 168
168	0 R 161	1 R 143	203	0 R 197	1 R 28
169	0 R 148	1 R 87	204	0 R 163	1 R 193
170	0 R 161	1 R 96	205	0 R 146	1 R 177
171	0 R 181	1 R 122	206	0 R 161	1 R 194
172	0 L 193	1 R 60	207	0 R 45	1 R 100
173	0 R 145	1 R 88	208	0 L 4	0 L 208
174	0 R 152	1 R 202			

B Exact identities and provenance

Final dense U209 table

`5ded628d692ff442e4346c61eb3db3c77b0ac2c80c1ecced45cbfc483be40e08`

Final native U209 table

`d0a155d3c683789f3bf5327c0cfb2a0cfd78076af8f1654d4ca426ccbb604d16`

Final source sites

`cddbed7f20e7a8b085b7ca0e76811d07b021522375643b5d1134eec0d2d23164`

Final reassociated pre-fold 212-state table

`1ca076433b5f4be053477fd2552eea9faa6542bb2405e33e43c46ced5e165c7a`

Common U212 source sites

`7b6539dc67b66ff73f22554018d7af5985d85574460a73fea5f025103c523d00`

Hashes identify exact reviewed artifacts; they do not prove the semantic properties. The checkers derive

their properties from those literal artifacts. The original machines, compiler, numerical calculus, theory arguments, and symbolic checker remain attributed to the supplied sources and their retained notices. This report contributes the compatible 209-state layout, its combined local proof, fresh final-table verification, comparison, and reproducible package.

References

[1] Vladimir Reshetnikov, ProveIt, pinned source snapshot, https://github.com/VladimirReshetnikov/ProveIt/tree/da4ab9dd68ccfc47ac852118601eaf1fdae53795/docs/incoming.

[2] *U211 ConPA Contextual*, supplied archive `U211_ConPA_Contextual_2026-10-28.zip`, retained under `reference/`.

[3] *U211 ConPA Research: Decoder Reassociation*, supplied archive `U211_ConPA_Research_2026-10-28.zip`, retained under `reference/`.

[4] *U211 ConPA Rotated*, supplied archive `U211_ConPA_Rotated_2026-10-28.zip`, retained under `reference/`.

[5] *Unary power-cover theory*, supplied `pa_unary_power_cover.zip`, including its full theory report and proof translations, retained under `reference/`.

[6] ProveIt, *Formal correctness of the exact 215-state machine*, https://github.com/VladimirReshetnikov/ProveIt/tree/main/Computability/BusyBeaver/Research/ZFC/k225-formal/verification/k215.

[7] Elliot Glazer, *K215 halts iff ZFC is inconsistent*, Metamath Zero formalization in PA, https://github.com/elliotglazer/k215-pa.

[8] Elliot Glazer, *k230-rocq: Theory K*, https://github.com/elliotglazer/k230-rocq#the-theory-k.

[9] Jeff87654, *A verified 254-state Turing machine that halts iff ZFC is inconsistent*, https://github.com/jeff87654/busy-beaver/tree/zf-254/ZF/m254.